

Developer Contribution Summary Report

Hospital Management System (HMS) Backend

This report outlines the core modules and key features developed for the Hospital Management System (HMS) backend.

1. Authentication & Security Module (auth)

Role: System foundation for security and access control.

- JWT Authentication: Implemented secure token-based login and token refreshing.
- Role-Based Access Control (RBAC): Middleware for role separation (super_admin, hospital_admin, doctor, patient, billing_admin).
- Multi-Tenancy Support: Data isolation using hospitalId to run multiple hospitals on one system.

2. Billing & Invoicing Module (billing)

Role: Handles hospital revenue and patient payment flow.

- Dynamic Calculations: Auto-calculates total amount, discount, tax/GST, and payableAmount.
- PDF Invoice Generation: On-the-fly professional PDF receipt generation using pdfkit.
- Smart Search Logic: Regex-based patient name search for quick billing.
- Cross-Module Sync: Automatically updates Appointment status when a bill is marked as paid.

3. Financial & Payroll Module (finance)

Role: Tracks overall hospital income and expenses.

- Expense Management: Tracks utility bills, equipment maintenance, and general expenses.
- Payroll System: Flow for staff salary, bonuses, and deductions.
- Profitability Tracking: Aggregates data to show net profitability of the hospital.

4. Doctor Management Module (doctor)

Role: Digitizes doctor working details.

- Profile Management: Manages qualifications, specialization, and experience.
- Availability & Fees: APIs to set working schedules and consultation fees.

5. Consultation Module (consultation)

Role: Digital record of doctor-patient checkups.

- Clinical Records: Securely saves and fetches diagnosis, symptoms, and medical notes.

6. Video Consultation Module (video-consultation)

Role: Enables telemedicine and remote health services.

- Meeting Link Generation: Dynamically generates unique video meeting links (e.g., Jitsi/WebRTC) for online appointments.

7. Laboratory Module (laboratory)

Role: Pathology and test reports management.

- Lab Requests Tracking: Tracks test status (Pending, Completed).
- Report Uploading: Uploads final test PDFs and links them to patient profiles.

8. Ward Management Module (ward)

Role: Hospital capacity and rooms management.

- Bed Availability Tracking: Real-time availability check for different ward types (ICU, General).

9. Inventory Management Module (inventory)

Role: Tracks hospital medical and general stock.

- Stock Tracking: Tracks current quantity of equipment and daily items.
- Supply Chain: Adds new items and deducts stock upon usage.

Key Architecture Highlights

"Implemented a Micro-service oriented approach. All 9 modules are loosely coupled but perfectly synchronized (e.g., Billing updates Appointment and Finance). Leveraged advanced Node.js capabilities including strict RBAC for security and dynamic PDF generation for user experience."